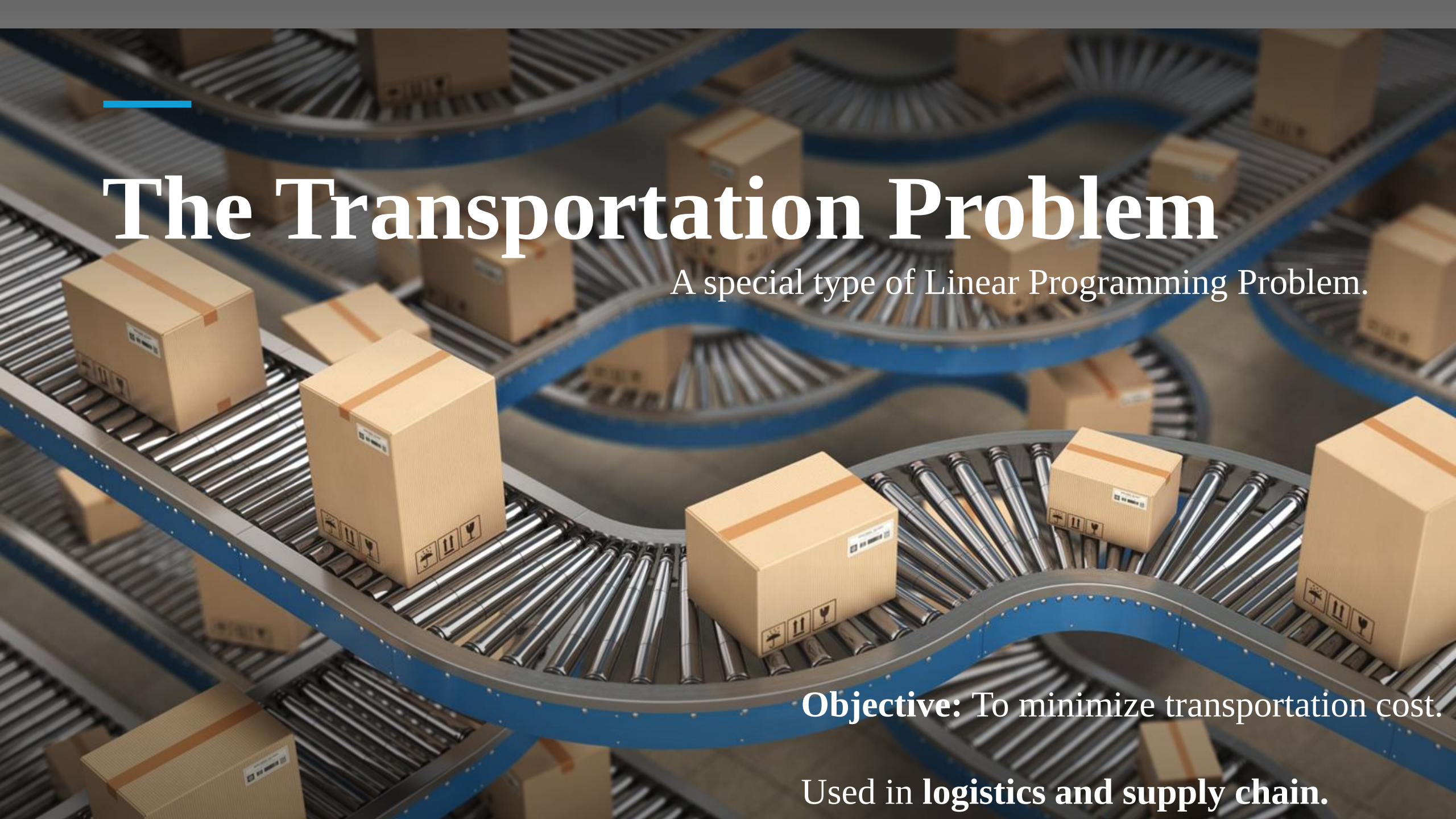


OPERATIONS RESEARCH



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The Transportation Problem

A special type of Linear Programming Problem.

Objective: To minimize transportation cost.

Used in **logistics and supply chain.**



What is a Transportation Problem?

A transportation problem deals with the distribution of a single homogeneous commodity from several sources (origins) to several destinations.

- There are multiple sources (origins) where goods are available.
- There are multiple destinations where goods are required.
- The cost of transportation from each source to each destination is known and may differ.
- The objective is to **minimize the total transportation cost** while satisfying all supply constraints and meeting all demand requirements.

Example

Consider the following situation:

- A fan manufacturing company has m factories located in n different cities.
- The fans produced in these factories are supplied to n retail dealers located in n different cities.

The transportation problem:

To form the transportation schedule that will minimize the total cost of transporting fans from different factory locations to different retail dealers.

Assumptions:

- There are m origins (sources) and n destinations (n may or may not be equal to m)
- Supply a product at i^{th} origin = a_i units
- Demand of the same at j^{th} destination = b_j units
- **Balanced Condition:** Total supply = Total demand ($\sum_{i=1}^m a_i = \sum_{j=1}^n b_j$)
- Cost of transporting 1 unit from i^{th} origin to j^{th} destination = c_{ij}
- Quantity transported from i^{th} origin to j^{th} destination = x_{ij}

Transportation Problem:

To determine non-negative values of x_{ij} satisfying the availability restrictions as well as the requirement restrictions, in such a way that the total transportation cost is minimized.

Mathematically:

$$\text{Min } Z = \sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij}$$

Subject to

- $x_{ij} \geq 0$
- $\sum_{j=1}^n x_{ij} = a_i, i = 1, 2, \dots, m$

(Supply Constraints/ Row sum Constraints :

Total goods sent from each origin = its supply)

- $\sum_{i=1}^m x_{ij} = b_j, j=1, 2, \dots, n$

(Demand Constraint/ Column sum Constraints:

Total goods received at each destination = its demand)

Relation Between Assignment and Transportation Problem:

- An assignment problem is a **simplified transportation problem**, where each origin is associated with one and only one destination.

Important Terminologies

- **Feasible solution:** A feasible solution to a transportation problem is a set of non-negative individual allocations (≥ 0) which satisfies the row and column sum restrictions.
- **Basic feasible solution (BFS):** A feasible solution of an m by n transportation problem is said to be basic if the total number of positive allocations is equal to $m + n - 1$ (One less than the sum of the number of rows and columns).
- **Optimum solution:** A feasible solution (not necessarily basic) is said to be optimum if it minimizes the total transportation cost.
- **Non-degenerate Basic Feasible Solution:** A BFS of a transportation problem is said to be non-degenerate BFS if the allocations are in independent position.

(Independent positions of a set of allocations means that it is always impossible to form any closed loop through these allocations.)

Solution of a Transportation Problem

The solution of a transportation problem consists of the two steps:

- **Step 1:** To find an initial basic feasible solution.
- **Step 2:** To obtain an optimal solution by making successive improvements to initial basic feasible solution (obtained in step 1) until no further decrease in the transportation cost is possible.

Methods to find an Initial Basic Feasible Solution:

- **Method 1:** North-West Corner Rule.
- **Method 2:** Lowest Cost Entry (Matrix Minima) Method.
- **Method 3:** Unit Cost Penalty Method (Vogel's Approximation Method).



North-West Corner Rule.

- In this method we always move to the right or down $\Rightarrow$ No loop can be formulated by drawing horizontal and vertical lines to the allocations.
- At each allocation at least one row or column is discarded from further consideration.
- The last allocation discards both a row and a column simultaneously.
- We cannot get more than $(m + n - 1)$ individual positive allocations $\Rightarrow$ We always get non-degenerate basic feasible solution.

Step 1:

- Start with the the top-most left corner, i.e., cell $(1, 1)$.
- Allocate there maximum possible amount.
- Thus, $x_{11} = \min(a_1, b_1)$.

Step 2:

Case I: If $b_1 < a_1$: $x_{11} = b_1$ and there is still some quantity available left in row 1.

- Move to the right-hand cell, i.e., $(1,2)$.
- Make the second allocation amount $x_{12} = \min(a_1 - x_{11}, b_2)$ in that cell.

Case II: If $b_1 > a_1$: $x_{11} = a_1$ and there is still some requirement left in column 1.

- Move vertically downwards to the next cel, i.e., $(2,1)$.
- Make the second allocation amount $x_{21} = \min(a_1, b_1 - x_{11})$ in this cell.

Case III: If $b_1 = a_1$: $x_{12} = 0$ or $x_{21} = 0$.

- Start from the new north-west corner of the transportation table.
- Allocate their as much as possible.

Step 3: Repeat steps 1 and 2 until all the available quantity is exhausted or all the requirement is satisfied.

Example

Find the initial basic feasible solution of the following transportation problem:

	W_1	W_2	W_3	Avail.
F_1	2	7	4	5
F_2	3	3	1	8
F_3	5	4	7	7
F_4	1	6	2	14
Req.	7	9	18	34

	W_1	W_2	W_3	(a_i)
F_1	5 (2)	(7)	(4)	5
F_2	2 (3)	6 (3)	(1)	8
F_3	(5)	3 (4)	4 (7)	7
F_4	(1)	(6)	14 (2)	14
(b_j)	7	9	18	34

- Multiplying each individual allocation by its corresponding unit cost and adding it we get the total cost corresponding to this feasible solution.

The feasible solution is :

$$5 * 2 + 2 * 3 + 6 * 3 + 3 * 4 + 4 * 7 + 14 * 2 = \text{Rs. 102.}$$

Lowest Cost Entry (Matrix Minima) Method.

NOTE: If the cell of lowest cost is not unique, we can select any one of these cells.

Step 1:

- Find the lowest cost in the cost matrix. Let it be c_{ij} .
- Allocate as much as possible in the cell (i, j) . $x_{ij} = \min(a_i, b_j)$

Step 2:

Case I: $x_{ij} = a_i$, i.e., the capacity of the i^{th} origin is completely exhausted.

- Cross out the i^{th} row of the transportation table
- Decrease the requirement b_j by a_i .
- Go to step 3.

Case II: $x_{ij} = b_j$, i.e., then the requirement of j^{th} destination is completely satisfied.

- Cross out the j^{th} column of the transportation table.
- Decrease a_i by b_j .
- Go to step 3.

Case III: $x_{ij} = a_i = b_j$

- Either cross-out the i^{th} row or j^{th} column but not both.
- Go to step 3.

Step 3:

- Repeat steps 1 and 2 for the reduced transportation table until all the available is exhausted or all the requirement is satisfied

Example

Find the initial basic feasible solution of the following transportation problem:

	W_1	W_2	W_3	Available
F_1	2	7	4	5
F_2	3	3	1	8
F_3	5	4	7	7
F_4	1	6	2	14
Requirements	7	9	18	34

	W_1	W_2	W_3	Available (a_i)
F_1	(2)	2 (7)	3 (4)	5 2 X
F_2	(3)	(3)	8 (1)	8 X
F_3	(5)	7 (4)	(7)	7
F_4	7 (1)	(6)	7 (2)	14 7 X
Requirements (b_j)	7 X	9 7 X	18 10 3 X	34

The transportation cost =

$$2 * 7 + 3 * 4 + 8 * 1 + 7 * 4 + 7 * 1 + 7 * 2 = \text{Rs. } 83$$



Unit Cost Penalty Method (Vogel's Approximation Method).

- Vogel's method usually takes more time as compared to other methods.
- However, it reduces the time in reaching the optimal solution.
- To obtain the optimal solution it is advised to find the initial BFS by Vogel's method.

Step 1:

- Identify the smallest and second smallest costs for each row of the transportation table.
- Find the difference between them for each row.
- Write these differences alongside the transportation table against the respective rows by enclosing them in parentheses. These are called '**penalties**'.
- Write the similar differences for each column below the corresponding column.

Step 2:

- Select the row or column for which the penalty is the largest.
- If a tie occurs, use any arbitrary tie breaking choice.
- Allocate the maximum possible amount to the cell with lowest cost in that particular row or column.
- Let the largest penalty correspond to i^{th} row and let c_{ij} be the smallest cost in the i^{th} row. Allocate the amount $x_{ij} = \min(a_i, b_j)$ in the cell (i, j) .
- Cross out the i^{th} row or the j^{th} column in the usual manner and construct the reduced matrix.

NOTE: If in the selected row or column the minimum cost is not unique, then locate in that cell in which more allocations can be made.

Step 3:

- Compute the row and column penalties for the reduced transportation table.
- Repeat the step 2.
- Continue this process until all the available quantity is exhausted or all the requirement is satisfied.

Example

Find the initial basic feasible solution of the following transportation problem:

	W_1	W_2	W_3	Available (a_i)
F_1	2	7	4	5
F_2	3	3	1	8
F_3	5	4	7	7
F_4	1	6	2	14
Requirements (b_j)	7	9	18	34

	W_1	W_2	W_3	(a_i)	Penalties
F_1	5 (2)	(7)	(4)	5 X	2 
F_2	(3)	(3)	(1)	8	2
F_3	(5)	(4)	(7)	7	1
F_4	(1)	(6)	(2)	14	1
(b_j)	7 2	9	18		
Penalties	1	1	1		

	W_1	W_2	W_3	(a_i)	Penalties
F_2	(3)	(3)	8 (1)	8 X	2 
F_3	(5)	(4)	(7)	7	1
F_4	(1)	(6)	(2)	14	1
(b_j)	2	9	18 10		
Penalties	2	1	1		

	W_1	W_2	W_3	(a_i)	Penalties
F_3	(5)	(4)	(7)	7	1
F_4	(1)	(6)	10 (2)	14 4	1
(b_j)	2	9	10 X		
Penalties	4	2	5 ↑		

	W_1	W_2	(a_i)	Penalties
F_3	(5)	7 (4)	7	1
F_4	2 (1)	2 (6)	4 2	5 ←
(b_j)	2 X	9 7		
Penalties	4	2		

Required BFS:

	W_1	W_2	W_3	(a_i)
F_1	5 (2)	(7)	(4)	5
F_2	(3)	(3)	8 (1)	8
F_3	(5)	7 (4)	(7)	7
F_4	2 (1)	2 (6)	10 (2)	14
(b_j)	7	9	18	

The total transportation cost =

$$5 * 2 + 8 * 1 + 7 * 4 + 2 * 1 + 2 * 6 + 10 * 2 = \text{Rs. } 80.$$

Loops and their Properties

Properties:

1. Every loop has an even number of cells.
2. A feasible solution to a transportation problem is basic iff, the corresponding cells in the transportation table do not form a loop.

Loop: An ordered set of four or more cells in a transportation table is said to form a loop if

- a) Any two adjacent cells of the set lie either in the same row or in the same column.
- b) No three or more adjacent cells lie in the same row or in the same column.

NOTE:

- The first cell of the set follows the last one in the set.
- If we join the cells of a loop by horizontal and vertical line segments we get a closed path.

Consider two sets:

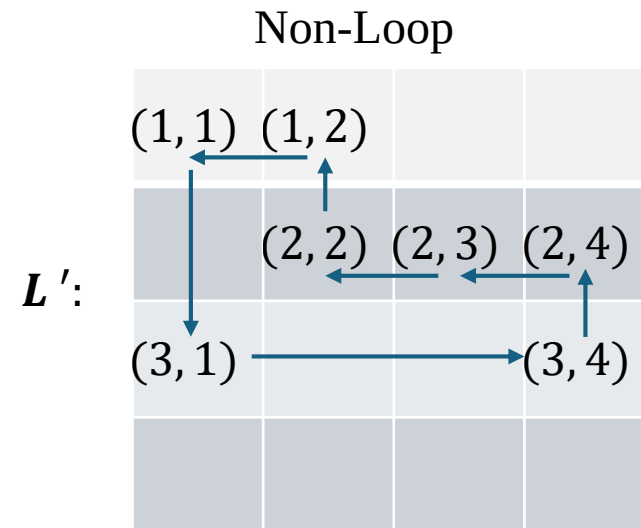
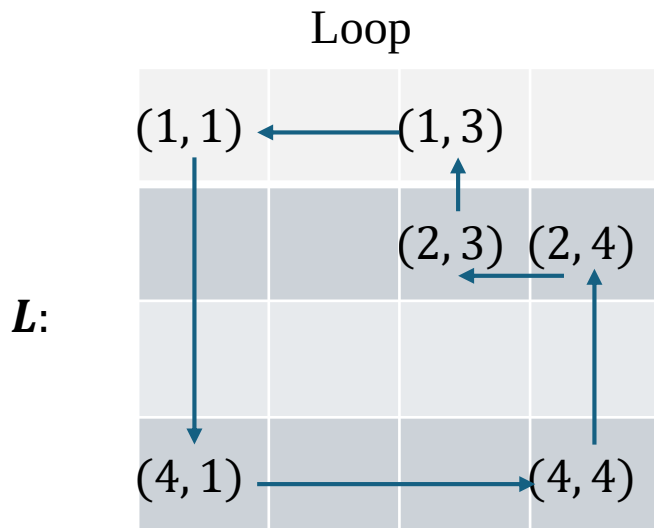
$$L = \{(1, 1), (4, 1), (4, 4), (2, 4), (2, 3), (1, 3)\}$$

and

$$L' = \{(3, 1), (3, 4), (2, 4), (2, 3), (2, 2), (1, 2), (1, 1)\}$$

where $(i, j)^{\text{th}}$ cell of the transportation table is denoted by (i, j) .

- The set L forms a loop.
- As the three cells $(2, 4)$, $(2, 3)$ and $(2, 2)$ lie in the same row, the set L' does not form a loop.



Optimality Test

- After obtaining an initial BFS to a given transportation problem, we need to test it for optimality, i.e., whether or not the solution minimizes the total transportation cost.
- Optimality test is **applicable to a non-degenerate BFS**, i.e., to a feasible solution consisting of $(m + n - 1)$ allocations in independent positions.

Computational Procedure of Optimality Test

Step1: Determine a set of $(m + n)$ numbers u_i ($i = 1, 2, \dots, m$) and v_j ($j = 1, 2, \dots, n$) such that for each **occupied** cell (r, s)

$$c_{rs} = u_r + v_s.$$

Step2: Compute the cell evaluation d_{ij} for each **unoccupied** cell (i, j) using the formula

$$d_{ij} = c_{ij} - (u_i + v_j).$$

Step3: Examine the matrix of cell evaluations as follows:

- a) If all $d_{ij} > 0$, then the solution under test is **optimal and unique**.
- b) If all $d_{ij} \geq 0$, with at least one $d_{ij} = 0$, then the solution under test is optimal and **alternative optimal solution** exists.
- c) If at least one $d_{ij} < 0$, then the solution under test is **not optimal**.

Transportation Algorithm for Minimization Problem (MODI Method)

Step I:

- Construct transportation table entering the capacities $a_1, a_2, \dots, a_m$ of the origins and the requirements $b_1, b_2, \dots, b_n$ of the destinations.
- Enter the costs c_{ij} in () at the upper left corners of all the cells.

Step II: Find an initial basic feasible solution.

Step III: Enter value of $u_i + v_j$ for each unoccupied cell (i, j) in the upper right corner of the corresponding cell, and d_{ij} at the lower right corner of the corresponding cell.

Step IV: Do the Optimality Test by evaluating sign of d_{ij} for each unoccupied cell (i, j) .

- If each $d_{ij} \geq 0$, we are done.
- Otherwise, go to **Step V**.

Step V: (When at least one $d_{ij} < 0$)

- Select the min (most negative) d_{ij} .
- Form a closed loop by joining this cell of min d_{ij} with the cells of positive allocations.
- Allocate to this cell as much as possible by vacating at least one of the pre-occupied cells and maintaining the row and column sum restrictions.
- This will give a new (improved) BFS.

Step VI:

- Repeat the Steps III to V to test the optimality of this new BFS.
- Continue the process until an optimum basic feasible solution is obtained.

Example 1

Determine the optimum basic feasible solution to the following transportation problem:

	D ₁	D ₂	D ₃	D ₄	a _i
O ₁	5	3	6	2	19
O ₂	4	7	9	1	37
O ₃	3	4	7	5	34
b _j	16	18	31	25	

Using Vogel's method, the initial BFS is obtained as

	D ₁	D ₂	D ₃	D ₄	a _i
O ₁	(5) 18	(3) 1	(6)	(2)	19
O ₂	(4) 12	(7)	(9)	(1) 25	37
O ₃	(3) 4	(4)	(7) 30	(5)	34
b _j	16	18	31	25	

Total transportation cost = 355.

- Since all rows contain the same (maximum) number of allocations, so take any of the $u_i = 0$.

- We take $\mathbf{u}_3 = \mathbf{0}$, which implies

$$\mathbf{v}_1 = 3 \text{ and } \mathbf{v}_3 = 7.$$

$$\mathbf{v}_1 = 3 \text{ implies } \mathbf{u}_2 = 1.$$

$$\mathbf{u}_2 = 1 \text{ implies } \mathbf{v}_4 = 0.$$

$$\mathbf{v}_3 = 7 \text{ implies } \mathbf{u}_1 = -1.$$

$$\mathbf{u}_1 = -1 \text{ implies } \mathbf{v}_2 = 4.$$

	D ₁	D ₂	D ₃	D ₄	u_i
O ₁	(5) 2 3	(3) 18	(6) 1	(2) -1 3	-1
O ₂	(4) 12	(7) 5 2	(9) 8 1	(1) 25	1
O ₃	(3) 4	(4) 4 0	(7) 30	(5) 0 5	0
v_j	3	4	7	0	

- Since all $d_{ij} \geq 0$, the solution under test is optimal.
- The solution of the given problem is:
 $x_{12} = 18, x_{13} = 1, x_{21} = 12, x_{24} = 25, x_{31} = 4, x_{33} = 30.$
- Minimum transportation cost = Rs. 355.
- As $d_{32} = 0$, an alternative optimal solution will also exist.

Example 2

Solve the following transportation problem in which cell entries represent unit cost:

	To			Available
	2	7	4	
From	2	7	4	5
	3	3	1	8
	5	4	7	7
	1	6	2	14
Required	7	9	18	

Using Vogel's method, the initial BFS is obtained as

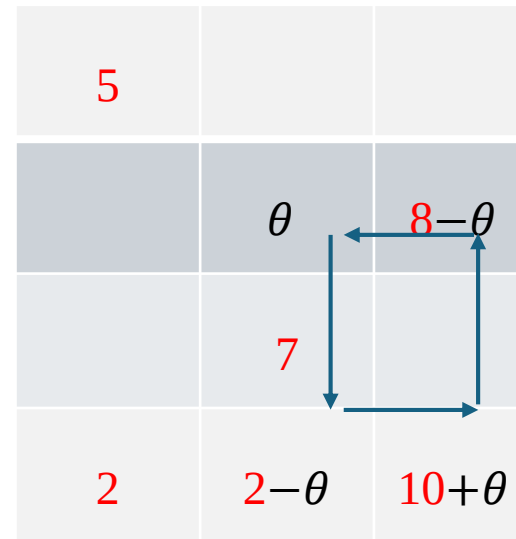
			a_i		
b_j	(2)	(7)	(4)	5 8 7 14	
	5				
	(3)	(3)	(1)		8
	(5)	(4)	(7)		
	(1)	(6)	(2)		7
	2	2	10	14	
	7	9	18		

Total transportation cost = Rs. 80.

- As row 4 has maximum allocation, choose $u_4 = 0$, and using costs of occupied cells calculate all u_i and v_j .
- Write $u_i + v_j$ in the upper right corner of each unoccupied cells.
- Evaluate d_{ij} for unoccupied cell and write at the lower right corner of the cell.

				u_i
(2)	(7)	7 (4)	3	
5		0	1	1
(3)	0 (3)	5 (1)		
	3	-2	8	-1
(5)	-1 (4)	(7)	0	
	6		7	-2
(1)	(6)	(2)		
				0
v_j	2	2	10	
	1	6	2	

- As $d_{22} = -2 < 0$, the solution under test is not optimal.
- The solution can be improved.
- Since minimum (max neg.) d_{ij} is $d_{22} = -2$, we allocate (say, θ) to the cell (2, 2) as much as possible.
- Identify the loop joining cell (2, 2) with the occupied cells.



Rule to determine θ :

- The value of θ is obtained by **equating** the minimum of the allocations **containing $-\theta$** at the corners of the closed loop **to zero**.
- Here, $\min [8 - \theta, 2 - \theta] = 0 \Rightarrow 2 - \theta = 0$ or **$\theta = 2$ units**.
- Thus, cell (2, 2) enters the solution, while cell (4, 2) leaves the solution.

Improved BFS is:

(2)	5	(7)	(4)
(3)		(3)	(1)
(5)		(4)	(7)
(1)	2	(6)	(2)
			10

Total transportation cost for this BFS = **76** < 80.

Optimality Test for new BFS:

				u_i
(2)	(7)	5 (4)	3	
5		2	1	1
(3)	0 (3)	(1)		
	3	2	8	-1
(5)	1 (4)	(7)	2	0
	4		5	
(1)	(6)	7 (2)		0
			2	
v_j	2	4	10	
	1		2	

- Since all $d_{ij} > 0$, the solution under test is optimal and exist uniquely.

Degeneracy in Transportation Problems

- A BFS to an m -origin and n -destination transportation problem can have at-most $(m + n - 1)$ independent non-zero allocations.
- If the number of allocation is exactly $(m + n - 1)$, the BFS is said to be non-degenerate otherwise it is said to be a degenerate one.
- Degeneracy in transportation problem can occur in two ways :
 - (i) BFS may be degenerate from the initial stage onward.
 - (ii) It may become degenerate at any intermediate stage, when the selection one entering cell empties two or more pre-occupied cells simultaneously.
- To resolve degeneracy, we allocate an extremely small amount Δ (close to zero) to one or more unoccupied cells of the matrix (generally lowest cost cells if possible), so that the total number of occupied cells become $(m + n - 1)$ at independent positions.
- Δ satisfies the following conditions:
 - 1) $\Delta < x_{ij}$ for $x_{ij} > 0$.
 - 2) $x_{ij} + \Delta = x_{ij} = x_{ij} - \Delta$ for $x_{ij} > 0$.
 - 3) $\Delta + 0 = \Delta$.
 - 4) If there are more than one Δ introduced in the solution :
 - If Δ, Δ' are in the same row :
 $\Delta < \Delta'$ when Δ is to the left to Δ' .
 - If Δ, Δ' are in the same column:
 $\Delta < \Delta'$ when Δ is above Δ' .
- Even after introducing Δ , the original solution of the problem is not changed.
- It is only a technique to apply the optimality test.
- As Δ has no physical significance, ultimately it is to be omitted.

Example

A manufacturer wants to ship 8 loads of his product as shown in the given table.

	D ₁	D ₂	D ₃	Available
O ₁	50	30	220	1
O ₂	90	45	170	3
O ₃	250	200	50	4
Required	4	2	2	

The matrix gives the mileage from origin O to destination D. Shipping costs are Rs. 10 per load per mile. What shipping schedule should be used

The initial BFS by Vogel's method is obtained as

	D ₁	D ₂	D ₃	a _i
O ₁	(50) 1	(30)	(220)	1
O ₂	(90) 3	(45)	(170)	3
O ₃	(250)	(200) 2	(50) 2	4
	b _j	4	2	2

- Since the total number of allocations is 4 which is one less than $m + n - 1 = 5$, this solution is a degenerate solution.
- To test the solution for optimality, we allocate Δ to the cell (1, 2) getting 5 allocations at independent positions.

	D ₁	D ₂	D ₃	u_i
O ₁	(50) 1	(30) Δ	(220) -120 340	-170
O ₂	(90) 3	(45) 70 -25	(170) -80 250	-130
O ₃	(250) 220 30	(200) 2	(50) 2	0
v_j	220	200	50	

- Since $d_{22} < 0$, so the solution under test is not optimal.
- We need to allocate to the cell (2, 2) as much as possible.
- Thus, we take Δ from cell (1, 2) to cell (2,2) and form the new table to check the solution for optimality.

	D ₁	D ₂	D ₃	u_i
O ₁	(50) 1	(30) 5 25	(220) -145 365	-195
O ₂	(90) 3	(45) Δ	(170) -105 275	-155
O ₃	(250) 245 5	(200) 2	(50) 2	0
v_j	245	200	50	

- Since for unoccupied cells all $d_{ij} > 0$, the solution under test is optimal.
- Thus, the solution of the given problem is

$$x_{11} = 1, \quad x_{21} = 3, \quad x_{32} = 2, \quad x_{33} = 2.$$
- Minimum mileage = $50 * 1 + 90 * 3 + 200 * 2 + 50 * 2 = \text{Rs. } 8200.$

Unbalanced Transportation Problems

- In a transportation problem, if the sum of all available quantities is not equal to the sum of all requirements, i.e., if

$$\sum_{i=1}^m a_i \neq \sum_{j=1}^n b_j ,$$

then such problem is called an unbalanced transportation problem.

- An unbalanced transportation problem may occur in two different forms :

(i) Shortage in Availability, i.e., $\sum_{i=1}^m a_i < \sum_{j=1}^n b_j$.

- To modify this type of unbalanced transportation problem to balanced type we introduce a dummy source row in the transportation table.
- The unit transportation costs from this dummy source to any destination are all set equal to zero.
- The availability at this dummy source is assumed to be equal to the difference $\sum_{j=1}^n b_j - \sum_{i=1}^m a_i$.

(ii) Excess of Availability, i.e., $\sum_{i=1}^m a_i > \sum_{j=1}^n b_j$.

- To modify this type of unbalanced transportation problem to balanced type we introduce a dummy destination column in the transportation table.
- The unit transportation costs to this dummy destination from any source are all set equal to zero.
- The requirement at this dummy destination is assumed to be equal to the difference $\sum_{i=1}^m a_i - \sum_{j=1}^n b_j$.

Example

Solve the following transportation problem:

	D_1	D_2	D_3	a_i
O_1	4	3	2	10
O_2	2	5	0	13
O_3	3	8	6	12
b_j	8	5	4	

- Since, $\sum_{i=1}^3 a_i = 35 \neq \sum_{j=1}^3 b_j = 17$, the problem is of unbalanced type.
- We convert this problem to a balanced one by introducing a dummy destination D_a with requirement $35 - 17 = 18$ having all the transportation costs equal to zero.
- The balanced transportation table is given as:

	D_1	D_2	D_3	D_a	a_i
O_1	4	3	2	0	10
O_2	2	5	0	0	13
O_3	3	8	6	0	12
b_j	8	5	4	18	

- Applying the Vogel's method in the usual manner, the initial BFS is obtained as

	D ₁	D ₂	D ₃	D _a	a _i
O ₁	(4) 5	(3) 5	(2)	(0) 5	10
O ₂	(2) 8	(5)	(0) 4	(0) 1	13
O ₃	(3)	(8)	(6)	(0) 12	12
b _j	8	5	4	18	

- This gives the transportation cost =

$$3 * 5 + 0 * 5 + 2 * 8 + 0 * 4 + 0 * 1 + 0 * 12 \\ = \text{Rs. } 31.$$

- Optimality test the solution:

	D ₁	D ₂	D ₃	D _a	u _i
O ₁	(4) 2 (3) 5 (2) 0 (0) 5				0
O ₂	(2) 8 (5) 3 (0) 4 (0) 1				0
O ₃	(3) 2 (8) 3 (6) 0 (0) 12				0
v _j	2	3	0	0	

- Since all $d_{ij} > 0$, so the solution under test is optimal.
- Thus, the solution of the given problem is

$$x_{12} = 5, \quad x_{31} = 8, \quad x_{23} = 4$$

and min. cost = Rs. 31.

(Note: Allocations in dummy column are not considered)